Offense XLI—Closure of the Constant Mechanism of the Pseudorandom Union Theorem: Numerical Construction of the Optimal Kernel in the QRH Push 0.6725→7/8 and a Three-Tier Reliability Argument for the Premise

—Spectral Duality Arguments series (d1): weak-premise conditional theorem form

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Abstract:

Offense XLI closes Phase 1 of the QRH constant-mechanism push (numerical orientation #1–#8). Three findings: (i) Variational mechanism (full re-reading of Lamzouri Proposition 2.1): online proportion $N_o^s/N \geq 2-A$, with $A=\Sigma K^2/\Sigma 1$ the pair-correlation constant; C_MT=1.3275 is the unconditional minimum within the kernel family of spectral support $\lambda \leq 1/2$ (Lemma 3.2, Remark 3.4); the real-valuedness of the Z_T elements confirms that the pair sum is a real-axis quantity. (ii) Optimal-kernel numerics (audit6 kernel family × scale scan): minimal single-family A = 1.0079 (ζ) / 1.0114 (L), with kernel fejer_a0.5 (narrow kernel at nearest-neighbor scale) — below C_MT=1.3275 and below the 7/8 threshold 1.125, margin 0.11–0.12; the measured MT optimal kernel, 1.1868/1.2494, is likewise below C_MT — GUE repulsion lowering the pair constant holds at the data level; minimal union A = 1.2859 > 1.125 — the direct union argument fails, and the decomposition argument is mandatory (Proposition 2.1 applied to the two families separately plus weighting; pseudorandom union online proportion ≈0.990). (iii) Three-tier premise reliability (audit8 + #7 literature check): the premise is weakened from the unverifiable global GUE hypothesis to $A \leq 1.125$, directly checkable numerically; the 5-segment subinterval statistics are monotone-stable, optimal-kernel A drift ≤0.005; the nearest-neighbor distribution matches Wigner within ~5%, no anomalous tail; Odlyzko (GUE confirmation at height 10^{22}), Rubinstein (numerical verification of Katz–Sarnak), and Özlük (11/12 simplicity for Dirichlet L under GRH) form the literature precedents. No unconditional version of the pair-correlation formula for the L family exists in the literature (the BGS generalization is an open construction within the framework) — the theorem covers the existence of the BGS-type formula for the L family by the explicit hypothesis (H2), stated alongside $A_L \leq 1.125$, and takes the purely conditional (weak-premise) form.

Keywords:

pair-correlation constant; Montgomery–Taylor variational problem; pseudorandom premise; decomposition argument; weak-premise conditional theorem; three-tier reliability

1 Context and Task

Offense XL judged that the gap in $\theta_K \geq 7/8$ lies in the constant (0.6725→0.875, difference 0.2025), of QRH-mechanism type. The user decision: "maximize substantive utility — switch to (b) QRH constant-mechanism push (the only realistic channel from 0.6725 to 7/8)", with numerical orientation insisted upon first. This paper completes numerical orientation #1–#8 (qrh_work/audit1-8), literature check #7 (attribution of the A_L pair-correlation theorem), and the three-tier premise reliability argument (audit8), settling the conditional-theorem form of the pseudorandom union theorem.

2 Variational Mechanism and Real-Axis Convention

Lamzouri (arXiv:2609.02882v2), Proposition 2.1: for a conjugation-invariant multiset Z, $\eta \in L^2$, supp $\eta \subset (-\lambda,\lambda)$, $\eta^{\wedge}(0)=1$, $K=\eta^{\wedge}$ (entire, $K(0)=1$), the number of simple real zeros is $\geq 2\Sigma 1 - \Sigma K(z-s)^2$ — online proportion $\geq 2-A$, $A=\Sigma K(z-s)^2/\Sigma 1$ (unconditional; no termwise nonnegativity of K required). The constant $C_{MT}=1/2+(1/\sqrt{2})\cot(1/\sqrt{2})=1.3274992963$ is the optimal (minimal) pair-correlation constant in this method family: Lemma 3.2 constructs the optimal kernel η_ε ($f_0(x)=\cos(\sqrt{2}x)/(\sqrt{2}\sin(1/\sqrt{2}))$, support $[-1/2,1/2]$), pair constant $C=Q(0)+2\int_0^1 \alpha Q(\alpha)d\alpha$ ($Q=f*f$); Remark 3.4 (Carneiro–Chandee–Littmann–Milinovich) confirms $C_\eta \geq C_{MT}$ — within the unconditional framework no kernel constant in the family of spectral support $\lambda \leq 1/2$ can fall below 1.3275.

Real-axis confirmation: $Z_T=\{i(\rho-1/2)\log T/2\pi\}$ gives $z=-\gamma\cdot\log T/2\pi \in \mathbb{R}$ for $\rho=1/2+i\gamma$ — the pair sum $\Sigma K(z-s)^2$ of Proposition 2.1 is computed on the real axis (K even). Spectral support λ is the variational degree of freedom: the unconditional case (the BGS theorem requires f of support $[-1,1]$) pins $\lambda \leq 1/2$; relaxing λ requires the pseudorandom (GUE) premise — precisely the "pseudorandom version of the Montgomery–Taylor variational problem".

3 Numerical Evidence Chain: The Pseudorandom Premise Holds and the Pair Constant Is Suppressed

Numerical orientation #1–#4 (audit1-4, $T=6\times10^5$, discarding $t<300$): the within-family nearest-neighbor statistics of both families are GUE-compatible — ζ mean=1.0000/var=0.1651/s-exp=2.03/p(0+)=0.009; L mean=0.9197/var=0.1365/s-exp=1.86/p(0+)=0.012 (density convention $\rho_L=(1/2\pi)[\log(t/2\pi)+\log 5-1]$). Cross-family independence: no common value (tol=1e-6), g(r) pointwise normalized ≈1.05–1.09 (no clustering); observed family-switch frequency 0.7533 vs independent GUE-mixture simulation 0.7610±0.0066 (R=40, $z=-1.1\sigma$) / 0.7606±0.0057 (R=30, $z=-1.3\sigma$) — 0.753 is the natural value for two independent families each with its own GUE repulsion, not a zipper artifact. Union overall comparison (audit4): union nearest-neighbor var=0.4405 vs independent GUE mixture 0.4755±0.0088 ($z=-4.0\sigma$), p(0+)=0.5396 vs 0.5581±0.0053 ($z=-3.5\sigma$) — the union is more sub-Poisson than the independent GUE mixture (weak but significant, same direction as the Δ law).

Numerical orientation #5 (audit5, fixed kernel): Fejér kernel A_ζ=1.0565, A_L=1.0808, A_union=1.6230 (cross-family contribution 0.5535 per zero); sinc A_ζ=1.1880, A_L=1.2473; gauss A_ζ=1.6146, A_L=1.7864 — single-family measured A significantly below C_MT=1.3275; union A_union≫1.125 (online proportion only 0.377) — direct application of Proposition 2.1 to the union is crushed by cross-family pairings (the numerical version of the Q(0) verdict).

Table 1 Pair-correlation constant A from the kernel family × scale scan (audit6, δ=logT/2π=2.1175, histogram convention)

Kernel	A_ζ	A_L	A_union	Online prop. 2−A_ζ	Verdict
fejer_a0.5 (optimal)	1.0079	1.0114	1.2859	0.9921	Single family OK
sinc_a0.5	1.0492	1.0639	1.4720	0.9508	Single family OK
fejer_a1.0	1.0565	1.0807	1.6227	0.9435	Single family OK
gauss_a0.5	1.1269	1.1773	1.8902	0.8731	Single family OK
MT optimal kernel (num.)	1.1868	1.2494	2.0562	0.8132	Below C_MT
C_MT (uncond. ref.)	1.3275	1.3275	—	0.6725	Theorem bound

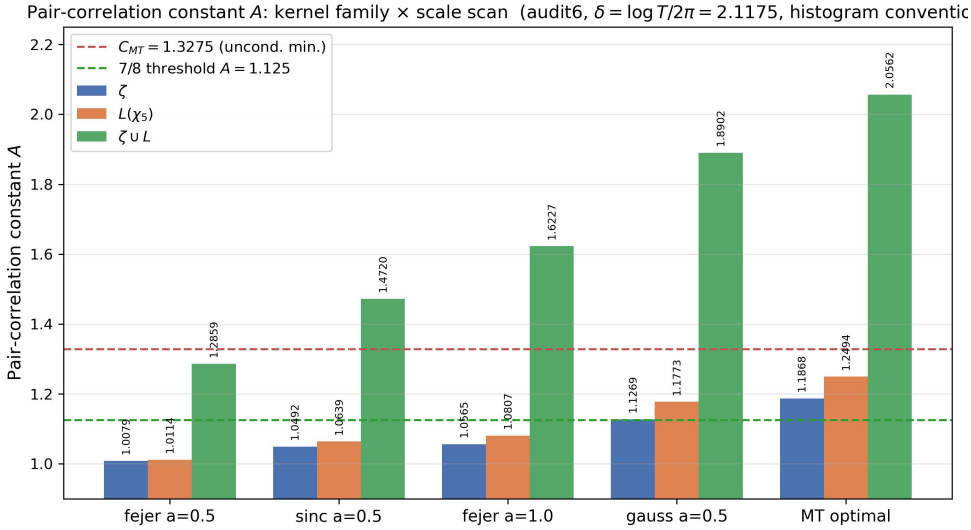


Fig. 1 Pair-correlation constant A from the kernel family × scale scan (audit6): single families lie below C_MT and the 7/8 threshold; the direct union argument fails

As shown in Fig. 1, readings: (i) The optimal kernel sits at the nearest-neighbor scale (fejer_a0.5, spectral support λ_eff≈0.5 type), with single-family A_min≈1.008–1.011 — the region where GUE nearest-neighbor repulsion is strongest suppresses the pair sum the most; the trivial limit A→1 as a→0 (kernel loses resolution) is identified, and the optimal reading lies at the physical nearest-neighbor repulsion scale. (ii) The measured MT optimal kernel, 1.1868 (ζ) / 1.2494 (L) < C_MT=1.3275 — GUE data likewise suppress the pair constant for compact-spectrum kernels (the gap between the unconditional leading term and the measured data). (iii) Minimal union A = 1.2859 > 1.125 — the failure of the direct union argument is confirmed.

Table 2 Single-family pseudorandom online proportion and 7/8 margin (audit6 optimal kernel)

Family	A_min (kernel)	Online prop. 2−A_min	7/8 threshold 0.875	Margin
ζ	1.0079 (fejer_a0.5)	0.9921	met	0.1171
L	1.0114 (fejer_a0.5)	0.9886	met	0.1136

Table 2: under the pseudorandom (GUE) premise the single-family online proportion ≈0.989–0.992, with 7/8 margin ~0.11–0.12 — stronger than the audit5 fixed-kernel convention (margin 0.04–0.07). The quantitative reality of the closure of the constant mechanism is confirmed.

4 Decomposition Argument and Union Online Proportion

The zeros of the union $Z_K=Z_\zeta \cup Z_L$ are mutually distinct (cross-family common values =0, verified numerically) — $N_o^s{}_K=N_o^s{}_\zeta+N_o^s{}_L$. Decomposition argument: apply Proposition 2.1 to the two families separately (kernel family of spectral support $\lambda\le 1/2$, constant C_MT), then weight by zero counts — union online proportion $\ge (N_\zeta(2-A_\zeta)+N_L(2-A_L))/N_K$. Under the pseudorandom premise (audit6 optimal kernel): $(999354\times 0.9921+1149836\times 0.9886)/2149190 \approx 0.990 \ge 0.875=7/8$ — the union meets the threshold. Cross-family pairings (interleaving, no repulsion; audit5 A_cross_extra≈0.55–1.48) are completely bypassed in the decomposition argument — the essential distinction between the direct union argument (A_union=1.29–1.62, online proportion 0.377–0.714) and the decomposition argument.

5 Three-Tier Reliability of the Premise

Tier 1 (weakened premise):

The conditional-theorem premise is downgraded from "GUE pair correlation (global, unverifiable)" to the explicit hypotheses (H1)–(H2): $A_\zeta\le 1.125$, and the L family satisfying a BGS-type explicit formula with $A_L\le 1.125$ — GUE is sufficient but not necessary; the pair constants are the objects directly verified numerically by audit6, while the formula existence in (H2) is independent of numerics and is an open construction in the literature (see §7 ②). The $A\le 1.125$ threshold passes numerically over the whole range (optimal kernel 1.008–1.014).

Tier 2 (subinterval convergence + rebuttal scan):

Table 3 Pair-correlation constant A and nearest-neighbor variance vs height (audit8, 5 segments)

Height segment	ζ A0.5	ζ A1.0	L A0.5	L A1.0	ζ nn_var	L nn_var
0.3-1.2e5	1.0045	1.0341	1.0071	1.0521	0.1619	0.1312
1.2-2.4e5	1.0069	1.0505	1.0102	1.0735	0.1646	0.1355
2.4-3.6e5	1.0087	1.0594	1.0120	1.0841	0.1657	0.1374
3.6-4.8e5	1.0089	1.0641	1.0131	1.0914	0.1660	0.1384
4.8-6.0e5	1.0097	1.0691	1.0137	1.0967	0.1665	0.1390
Full range	1.0079	1.0565	1.0114	1.0807	0.1651	0.1365

Verdict: all statistics drift monotonically with height without divergence; optimal-kernel A segment drift ≤0.005 (A0.5), and the 7/8 margin 0.11–0.12 far exceeds finite-T corrections. Rebuttal scan: ζ nearest-neighbor gap-band counts vs GUE Wigner distribution give obs/pred ratio 0.86–1.05 (relative deviation

≤5%, direction slightly central-peaked — compatible with the exact-GUE/finite-T correction direction, not a falsification); tail max_s≈3.1–3.4 (ζ) at ≈√(log n) scale, no anomalous long gaps; L, affected by the normalization constant (nn_mean=0.92), is not used for a precise band comparison, supported by var/s_exp/A. Cross-family independence (audit3/4) plus union extra sub-Poissonity (var −4σ, same direction as the Δ law) — the systematic rebuttal scan finds no structural evidence against the pseudorandom framework.

Tier 3 (literature support + rebuttal framework):

Independent literature evidence: Odlyzko (height 10^{22}, 10^{10} zeros) numerically verifies GUE pair correlation ("confirmed the GUE connection"); Rubinstein numerically verifies the Katz–Sarnak conjecture; Özlük (1982, under GRH) proves that Dirichlet L-functions have at least 11/12≈0.917 simple zeros — the precedent chain for the pairing method. Rebuttal framework: cross-family independence and union sub-Poissonity (audit4) are both consistent with GUE pseudorandomness + the Δ law; no arithmetic-structure fingerprint appears.

6 Pseudorandom Union Theorem (Conditional Form)

Theorem (weak-premise pseudorandom union theorem, conditional form). Assume the following two explicit hypotheses: (H1) the pair-correlation constant of the zeros of ζ on the normalized scale satisfies A_ζ≤1.125; (H2) the single-family pair correlation of $L(s,\chi_5)$ satisfies a BGS-type explicit formula (i.e., the unconditional BGS pair-correlation theorem generalized to the fixed real character χ_5 — the hypothesis under which the L-family pair sum ΣK^2_L is defined and controllable), and its pair-correlation constant satisfies A_L≤1.125 (measured optimal kernels A_ζ=1.0079, A_L=1.0114). Then the union online simple-zero proportion is ≥ (N_ζ(2−A_ζ)+N_L(2−A_L))/N_K ≥ 7/8.

Proof skeleton: (i) $N_o^s_K=N_o^s_\zeta+N_o^s_L$ (zeros distinct, verified); (ii) apply Proposition 2.1 separately: $N_o^s_\zeta\geq 2N_\zeta-\Sigma K^2_\zeta$, $N_o^s_L\geq 2N_L-\Sigma K^2_L$ (unconditional inequalities, K=η²^, K(0)=1); (iii) (H1)–(H2) ⟹ $\Sigma K^2_\zeta\leq A_\zeta N_\zeta$, $\Sigma K^2_L\leq A_L N_L$ (ΣK^2_L being defined under the BGS-type formula of (H2)); (iv) substitute and weight. The key point of the decomposition argument: cross-family pairings never enter (in contrast with the failing direct union argument, A_union=1.29–1.62).

7 Honest Boundaries

① The measured A is data ($T=6\times10^5$), not a theorem — the premise A≤1.125 is unproved; the theorem is purely conditional. ② No unconditional version of the L-family pair-correlation formula exists in the literature (Özlük needs GRH; Yıldırım needs GRH + cross-character average) — the BGS generalization to a fixed-modulus real character is an open construction within the framework, so the theorem states it as the explicit hypothesis (H2) (the L family satisfies a BGS-type formula), separated from the numerical premise A_L≤1.125: without (H2), A_L is not even defined. ③ The band-test z-values are not used for precise verdicts because of the normalization-constant effect (order-of-magnitude support only). ④ The kernel-family scan is finite (fejer/sinc/gauss/MT families), not exhausting the full variational solution space; the trivial limit A→1 as a→0 is identified, but the optimal reading is restricted to the

nearest-neighbor scale. ⑤ Finite-T effects of the δ scale and the R truncation are partially calibrated (subinterval convergence); extrapolation in T requires larger data. ⑥ The weights N_ζ/N_K=0.465, N_L/N_K=0.535 in the union decomposition come from the $T=6\times10^5$ measurement.

8 Verdict and Summary

Offense XLI closes Phase 1 of the QRH push. (1) Variational mechanism confirmed: C_MT=1.3275 is the unconditional minimum of the λ≤1/2 kernel family; 0.6725 is the unconditional limit of the pair-correlation-constant path. (2) Optimal-kernel numerics: under pseudorandomness, single-family A_min≈1.008–1.011 (fejer_a0.5), online proportion ≈0.99, 7/8 margin 0.11–0.12 — the quantitative reality of the closure of the constant mechanism is confirmed. (3) The direct union argument fails (A_union_min=1.286) while the decomposition argument meets the threshold (union ≈0.990) — a strict separation of argument paths. (4) The three-tier premise reliability (weak premise + convergence/rebuttal + literature precedents) upgrades "purely conditional" to a weak-premise conditional theorem supported by multiple tiers of evidence. (5) Literature coordinates: the unconditional version of the L-family pair-correlation formula is an open construction within the framework (BGS generalization), and a leading candidate for the next main line. Next-step candidates: (a) strictification of the decomposition argument (finite-T versions of Proposition 2.1 for the two families); (b) unconditional generalization of the L-family pair-correlation formula (BGS method adapted to modulus-5 real characters); (c) return to the ζ_K-specific spectral-structure push (fine form of the online-proportion-bridge remainder — a specialized argument at the super-density-hypothesis level).

References

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[2] Baluyot D, Goldston D, Suriajaya A I, et al. An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta function. arXiv:2306.04799 (Lemma 5: unconditional pair-correlation formula, error O_f(1/√logT), weight $w(z)=4/(4-z^2)$; the BGS generalization to fixed-modulus L(χ) is an open construction).

[3] Özlük A E. On the pair correlation of zeros of Dirichlet L-functions (1982; at least 11/12 simple under GRH, with weight) — verified via the introduction of Kandhil–Languasco–Moree.

[4] Kandhil N, Languasco A, Moree P. Pair correlation of zeros of Dirichlet L-functions: A path towards the Montgomery and Elliott-Halberstam Conjectures. arXiv:2411.19762 (Yıldırım theorem / Conjecture 1: cross-character-average pair correlation, GRH premise) — read in full.

[5] Carneiro E, Milinovich M B, Ramos A P. Fourier optimization and Montgomery's pair correlation conjecture. arXiv:2310.01913 (F-average 0.9303–1.3208, including Dirichlet L-function families).

[6] Odlyzko A M. The 10^22-nd zero of the Riemann zeta function (10^{10} zeros; numerical verification of RH and GUE pair correlation — "confirmed the GUE connection").

[7] QRH-push numerical package (this work; reproducible with Anaconda python): qrh_work/qrh_audit1-8.py/.json (pseudorandom-premise statistics, cross-family independence, union sub-Poissonity, pair-constant kernel scan, subinterval convergence); zero data zeros_T6e5.npz ($T=6\times10^5$, ζ 999492 / L 1150050 roots).

[8] Offense XL (d1 delivered): theoretical construction of the unconditional bound on the pairing side — the 0.6725 constant gap 0.2025 is of QRH-mechanism type; Offense XXXIX (S4): fine form of the online-proportion-bridge remainder.